Automated Weld Seam Detection and 3D Map Integration for Industrial Vessel Inspection Using Depth Cameras and 3D LiDAR

Nontanut Chitsiriparat[1,†], Run Janna[1,†], Theerawath Phetpoon[1], Wasuthorn Ausrivong[1], Kanut Tarapongnivat[1], Worameth Nantareekurn[1], Suppachai Pewkliang[2] and Poramate Manoonpong[1,*]

***Abstract*—The inspection of oil and gas pressure vessels is dangerous and difficult due to confined environments and strict safety constraints. While robots reduce human exposure, most existing inspection systems rely on image-based assessment and lack spatially consistent weld seam referencing for maintenance integration. This paper presents an integrated pipeline that combines deterministic depth-based seam extraction with LiDAR–IMU odometry for globally referenced weld seam localization. Seam centerlines are estimated from RGB-D depth profiles via plane alignment, polynomial detrending, and adaptive matched filtering, and the detected seam offsets are fused with FAST-LIO2 (Fast Direct LiDAR-Inertial Odometry) pose estimation to generate globally referenced seam trajectories without modifying the underlying odometry. Validation on longitudinal and circumferential seams shows per-frame centerline mean absolute errors (MAEs) of 0.76 mm and 1.10 mm, respectively; the global seam trajectory MAE is 13.37 mm (P95 = 26.06 mm). Along-seam target-point localization (simulated defects using black markers) achieves 2.60 cm MAE with 96.7% ordering consistency across 30 cm ground-truth intervals. The resulting seam-consistent annotation supports target-point indexing in global coordinates and direct registration with LiDAR maps or CAD models, enabling inspection-to-maintenance digital integration.**

***Index Terms*—Industrial inspection robotics, weld seam detection, target-point localization, RGB-D sensing, LiDAR–inertial odometry, seam trajectory reconstruction.**

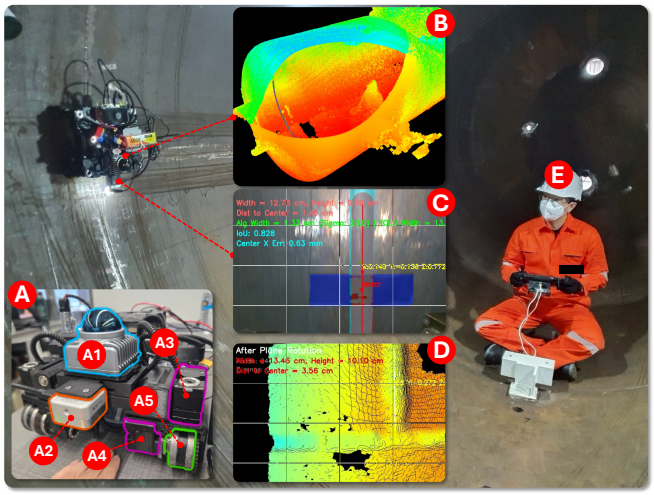


Fig. 1. **System overview and performance.** (**A**) Mobile inspection robot setup. (**A1**) and (**A2**) The 3D-LiDAR and RGB-D cameras, respectively. (**A3**) and (**A4**) The steering and driving motors. (**A5**) The magnetic wheel. (**B**) 3D point cloud of the vessel captured by the 3D-LiDAR sensor. (**C**)–(**D**) RGB image and 3D point cloud of the weld seam generated from the RGB-D camera, respectively. (**E**) The operator remotely controls the robot.

I. Introduction

Pressure vessels are critical assets in the petrochemical, power generation, and oil and gas industries. Weld seams are among the most failure-prone regions due to stress concentration, fatigue loading, and corrosion, requiring periodic inspection in accordance with API 510 [1].

Conventional inspection findings—narrative descriptions, drawings, and photographs—lack precise spatial references, complicating defect re-localization across inspection and repair visits and forcing repeated confined-space entries that extend outages and increase costs [2].

Robotic inspection offers an alternative to manual entry. Magnetic wall-climbing robots have been applied to seam inspection on ferromagnetic structures [3], while vision-based approaches focus on weld seam detection and tracking [2, 4]. LiDAR–inertial odometry systems such as FAST-LIO2 [5] and LIO-SAM [6] enable real-time pose estimation in GPS-denied environments, while digital twin platforms support three-dimensional visualization of spatially registered inspection data [7, 8]. Wall-climbing robotic weld inspection systems have also been demonstrated [9–13]. However, they localize seams in image space or robot-centric coordinates without a persistent global frame that is revisitable across inspection campaigns.

More broadly, these seam detection, localization, and visualization systems are typically developed in isolation: seam detection operates in a local sensor frame, simultaneous localization and mapping (SLAM) systems reconstruct geometry without seam-level semantics, and digital twin workflows

[1]These authors are with the School of Information Science and Technology (IST), Vidyasirimedhi Institute of Science and Technology (VISTEC), Rayong, Thailand.

[2]This author is with PTT Exploration and Production Public Company Limited, Bangkok, Thailand.

[†]These authors contributed equally to the work.

[*]Poramate Manoonpong is the corresponding author and can be contacted at poramate.m@vistec.ac.th

assume spatially registered measurements that most inspection systems do not provide, leaving inspection findings and maintenance planning loosely coupled.

Thus, this work proposes an integrated software framework coupling depth-based weld seam detection with LiDAR–inertial odometry on a magnetic-wheel inspection robot. We focus on geometric seam localization—target detection and classification are separate downstream tasks—with the objective of consistent spatial referencing for inspection and maintenance planning rather than metrology-grade reconstruction. The seam position is estimated as a lateral offset in the robot body frame and fused with LiDAR-based pose estimation to produce world-referenced seam trajectories. The main contributions are:

- A geometry-based weld seam detection method leveraging depth discontinuities and surface-normal variations to localize seams without prior CAD models or learned features.
- A synthetic seam point cloud approximating physical seam width along the reconstructed path, enabling intuitive visualization and registration with the as-built vessel geometry.
- A target-point localization strategy that maps detected seam points into the global frame via LiDAR–inertial odometry, providing actionable waypoints for downstream robotic welding in maintenance or inspection tasks.

Taken together, while these components are developed based on established techniques, to our knowledge, this is the first framework to integrate per-frame, depth-based seam measurements with LiDAR–inertial odometry. This integration provides globally referenced, revisitable seam trajectories, which are validated on an industrial vessel.

II. Mobile Inspection Robot Hardware

The proposed mobile inspection robot consists of four magnetic wheels and eight actuators in a swerve-drive configuration, enabling omnidirectional motion over ferrous vessel surfaces with stable adhesion and maneuverability on both longitudinal and circumferential weld seams. A 3D LiDAR sensor (Livox MID-360, Fig. 1-A1) and an RGB-D camera (Intel RealSense D405, Fig. 1-A2) are rigidly mounted on the robot's top plate: the LiDAR–inertial measurement unit (IMU) provides real-time odometry for global pose estimation, while the RGB-D camera captures high-resolution local 3D measurements of weld seam geometry. These systems operate on an NVIDIA Jetson Orin NX 16GB, running the Robot Operating System (ROS).

The swerve configuration integrates steering (Fig. 1-A3) and driving (Fig. 1-A4) motors, while passive joints in the wing structure allow the magnetic wheels (Fig. 1-A5) to conform to the surface curvature and maintain alignment. The ESP32-WROOM microcontroller sends motor commands and receives sensory feedback—steering angular position, driving angular velocity, and motor torque—via Micro-ROS over a wireless connection. The Jetson runs all perception, odometry, and mapping onboard; the ESP32 handles only low-level motor control and telemetry.

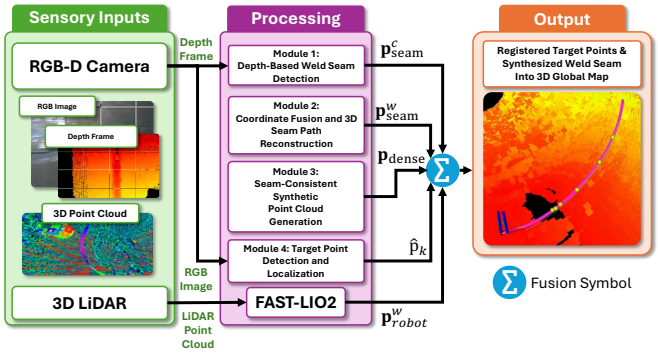


Fig. 2. **Data flow diagram of the proposed software system for automated weld seam detection and 3D map integration for pressure vessel inspection.** RGB-D depth data is processed for weld seam detection, while LiDAR–IMU odometry provides global pose estimation. Detected seam offsets are fused with robot poses to reconstruct a globally referenced seam trajectory.

III. System Architecture for Automated Weld Seam Detection and 3D Map Integration

The proposed system architecture detects weld seams and target points, annotating them directly onto a global 3D map built by FAST-LIO2 [5]; its tightly coupled LiDAR-inertial odometry is hereafter referred to as robot odometry. As shown in Fig. 2, the system comprises four sequential modules:

1) **Depth-based weld seam detection**: A depth profile from each RGB-D frame yields the lateral displacement of the seam center relative to the camera's optical axis and an approximate seam width.
2) **Coordinate fusion and seam path reconstruction**: Per-frame lateral seam center offsets are fused with robot odometry to transform local detections into a consistent seam path in the global map frame.
3) **Seam-consistent point cloud generation**: The reconstructed seam path produces a 3D seam representation overlaid onto the global map for visualization and downstream analysis.
4) **Target-point detection and localization**: Candidate target points along the reconstructed seam path are identified and projected onto the global map for inspection support.

A. Module 1: Depth-Based Weld Seam Detection

The first module extracts the weld seam centerline from each RGB-D frame via geometric alignment, row-wise signal filtering, and line fitting (Fig. 3). Industrial vessel surfaces are generally curved and arbitrarily tilted relative to the camera, so directly thresholding the raw depth is insufficient: the highest region often corresponds to the elevated portion of the

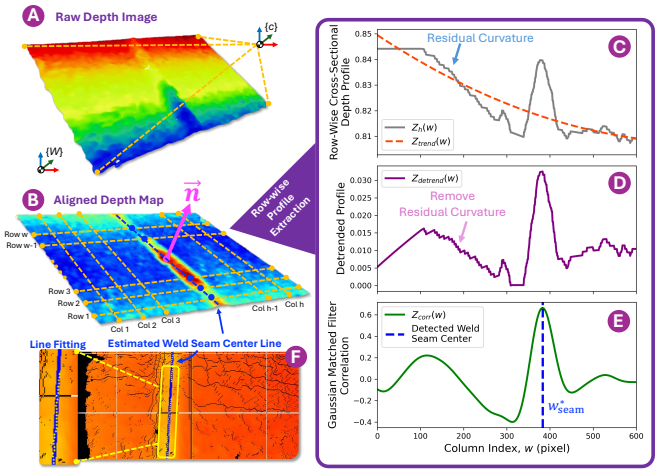


Fig. 3. **Depth-based weld seam detection.** (**A**) Raw depth image from the RGB-D camera, showing the weld seam as a surface depression across a curved vessel wall. (**B**) Aligned depth map after RANSAC plane fitting and Rodrigues' rotation, isolating the seam profile from the surface curvature, represented in the red region. (**C**) Row-wise cross-sectional depth profile showing the raw depth profile (gray) with a polynomial trend (orange). (**D**) Detrended profile showing the detrended response, where the seam appears as a distinct peak. (**E**) Gaussian matched- filter correlation showing the matched-filter response (green) and the detected weld-seam center index, w^*_{seam} (blue). (**F**) Line-fitting result overlaid on the aligned depth map, showing the detected seam centerline (blue).

curved surface rather than the seam (Fig. 3-A). A two-step geometric alignment therefore first estimates the dominant surface plane, then rotates the point cloud to align with it.

Plane Estimation. The pressure vessel surface is estimated as the dominant surface plane using RANSAC [14], maximizing the inlier count within a 0.01 m distance threshold over 200 iterations. The resulting plane $ax + by + cz + d = 0$ yields the surface normal $\vec{\mathbf{n}} = [a, b, c]^T$.

Tilt Compensation. The point cloud is rotated via Rodrigues' formula [15] so that the plane normal $\hat{\mathbf{n}}$ aligns with the camera z-axis $\vec{\mathbf{k}} = [0, 0, 1]^T$, with rotation axis $\mathbf{u} = (\hat{\mathbf{n}} \times \vec{\mathbf{k}})/\|\hat{\mathbf{n}} \times \vec{\mathbf{k}}\|$ and angle θ given by $\sin\theta = \|\hat{\mathbf{n}} \times \vec{\mathbf{k}}\|$, $\cos\theta = \hat{\mathbf{n}}^T\vec{\mathbf{k}}$. The rotation matrix is:

$$\mathbf{R}_{\text{align}} = \mathbf{I} + \sin\theta\,[\mathbf{u}]_\times + (1 - \cos\theta)\,[\mathbf{u}]_\times^2, \tag{1}$$

Each point $\mathbf{p} = [x, y, z]^T$ in the original point cloud is then transformed as $\mathbf{p}' = \mathbf{R}_{\text{align}}\,\mathbf{p}$. Since surface orientation varies gradually, the rotation is cached and updated every five frames.

This alignment enables height measurement relative to the vessel surface (Fig. 3-B), so that the most prominent elevated region corresponds to the weld seam. Two challenges remain: the seam may appear at an arbitrary orientation in the aligned depth map, and residual lateral (row-wise) curvature may cause simple thresholding to select the higher side of the surface. A three-stage signal-processing pipeline addresses both.

Row-wise Profile Extraction. The aligned depth map $Z_{\text{map}}(h_i, w)$ is constructed by projecting the rotated 3-D point cloud onto the image plane and aggregating multiple projected points per pixel via depth averaging. For each selected image row h_i, a one-dimensional cross-sectional profile is defined as:

$$Z_{h_i}(w) = Z_{\text{map}}(h_i, w), \tag{2}$$

where w denotes the horizontal pixel coordinate. Profiles are extracted with a vertical stride of two rows ($\approx$ 240 cross-sections for a 480-row image); missing depth values are filled by linear interpolation along w, and each cross-section is processed independently.

Polynomial Detrending. A second-order polynomial trend $\hat{Z}_{\text{poly}}(w) = \hat{a}_2 w^2 + \hat{a}_1 w + \hat{a}_0$ is fitted via ordinary least squares and subtracted:

$$Z_{\text{detrend}}(w) = Z_{h_i}(w) - \hat{Z}_{\text{poly}}(w). \tag{3}$$

After plane alignment and detrending, the weld seam appears as a distinct positive peak in $Z_{\text{detrend}}(w)$ (Fig. 3-C,D).

The RANSAC plane serves as a first-order normalization, not a planar surface assumption. At the short working distance ($\approx$ 0.08–0.10 m), the sagitta of the curved surface relative to its best-fit plane is on millimeter-scale, and this residual curvature is subsequently removed by the detrending in Eq. (3), yielding a local quadratic surface approximation. Consistent with this, the circumferential MAE (1.10 mm, Table I) remains close to that of the longitudinal case.

Gaussian Matched Filtering. To suppress residual low-frequency noise, the detrended profile is convolved with a unit-peak zero-mean Gaussian kernel [16]:

$$\begin{aligned} Z_{\text{corr}}(w) &= Z_{\text{detrend}}(w) * G(\xi), \\ G(\xi) &= e^{-\frac{\xi^2}{2\sigma^2}}, \quad \xi \in [-3\sigma, 3\sigma], \end{aligned} \tag{4}$$

where σ modulates the kernel width and $*$ denotes convolution. The seam center is $w^*_{\text{seam}} = \arg\max_w Z_{\text{corr}}(w)$. Sub-pixel refinement fits a quadratic $z(w) = \alpha_p w^2 + \beta_p w + \gamma_p$ to the three samples surrounding the discrete maximum, giving the refined peak location

$$w_{\text{seam}} = -\frac{\beta_p}{2\alpha_p}, \tag{5}$$

accepted when $\alpha_p < 0$ (concave peak) and the shift from the discrete maximum is strictly below 1.0 pixel (Fig. 3-E). The physical seam width Δw_{seam} is estimated from the full width at half maximum (FWHM) of $Z_{\text{detrend}}(w)$.

Adaptive Kernel Update. Rather than using a fixed σ, the filter parameter is updated online via temporal feedback: the average FWHM across all detected profiles in the k-th frame,

TABLE I
PER-FRAME LOCALIZATION BY SEAM GEOMETRY

Geometry	MAE (mm)	P95 (mm)	Angular Error (°)	F1	Cov. (%)
Longitudinal	**0.76**	**1.66**	1.61	**0.676**	83.5
Circumferential	1.10	2.06	**0.90**	0.650	**87.8**

$\overline{\mathrm{FWHM}}_k$, updates the kernel width for the subsequent frame by first-order exponential smoothing:

$$\sigma_k = (1-\alpha)\,\sigma_{k-1} + \alpha\,\frac{\overline{\mathrm{FWHM}}_k}{2.355}, \tag{6}$$

where $\alpha = 0.1$ (effective window ≈ 10 frames) and $2.355 = 2\sqrt{2\ln 2}$ is the FWHM-to-σ relation of a Gaussian, matching the kernel to the observed profile shape [16]. This causal update allows the seam width estimate in frame k to influence filtering in frame $k{+}1$.

Since weld seam geometry is manufactured within standardized tolerances, the seam width remains approximately consistent across spatial locations and frames. The exponential smoothing stabilizes σ against transient noise while adapting to gradual variations, and σ is clamped to $[3, 20]$ pixels for robustness.

The 1-D peak model targets centerline localization rather than full bead characterization. First-order width variation is captured via the adaptive kernel in Eq. (6), while richer profile models can also be applied in future work.

2-D Line Fitting. Applying the row-wise pipeline to the entire aligned depth map yields candidate seam center pixels $\mathbf{p}_{\mathrm{seam}} = (w_{\mathrm{seam}},\, h)$, one per processed row. Since individual rows may produce spurious detections due to surface irregularities or depth dropouts, the candidates are refined in two stages: RANSAC estimates a dominant 2-D line and rejects outliers, and the surviving inliers are refined via Total Least Squares (TLS), where the first right singular vector of the zero-mean inlier coordinates, computed by singular value decomposition (SVD), defines the fitted line direction (Fig. 3-F).

Back-Projection to 3-D. Each 2-D inlier is lifted into three-dimensional space using the corresponding aligned depth value and the camera intrinsic parameters:

$$\mathbf{p}^{c}_{\mathrm{seam}} = Z_{\mathrm{map}}(w,h)\begin{bmatrix}(w-c_x)/f_x\\(h-c_y)/f_y\\1\end{bmatrix}, \tag{7}$$

where $\mathbf{p}^{c}_{\mathrm{seam}} \in \mathbb{R}^3$ is the seam center point in the camera frame $\{c\}$, f_x, f_y are the focal lengths, and c_x, c_y are the principal point coordinates.

3-D Line Refinement. To enforce structural consistency and suppress residual per-frame noise, an SVD-based TLS line is fitted to the back-projected points, imposing a straight-line prior consistent with longitudinal weld seams and improving geometric stability before coordinate fusion. The geometric centroid of the 3-D inlier set, denoted $\bar{\mathbf{p}}^{c}_{\mathrm{seam}}$, is taken as the single representative per-frame seam center transmitted to Module 2 (described below). The fitted seam direction vector is retained as an auxiliary output for downstream robotic welding applications, whereas the coordinate fusion in Module 2 operates on $\bar{\mathbf{p}}^{c}_{\mathrm{seam}}$ alone.

B. Module 2: Coordinate Fusion and 3D Seam Path Reconstruction

This module transforms the per-frame seam center $\bar{\mathbf{p}}^{c}_{\mathrm{seam}}$ from the camera frame $\{c\}$ into the global frame $\{W\}$ provided by FAST-LIO2 via two chained rigid-body transformations. The camera-to-body transform $(\mathbf{R}^{b}_{c}, \mathbf{t}^{b}_{c})$ is determined from rigid mechanical mounting geometry; due to the fixed bracket and co-aligned optical axis, $\mathbf{R}^{b}_{c}$ is close to the identity matrix, with residual rotational misalignment bounded by manufacturing tolerances (estimated $< 2^\circ$). The body-to-world transform $(\mathbf{R}^{W}_{b}, \mathbf{t}^{W}_{b})$ is estimated online using FAST-LIO2. The globally referenced seam point is computed as:

$$\mathbf{p}^{W}_{\mathrm{seam}} = \mathbf{R}^{W}_{b}\left(\mathbf{R}^{b}_{c}\,\bar{\mathbf{p}}^{c}_{\mathrm{seam}} + \mathbf{t}^{b}_{c}\right) + \mathbf{t}^{W}_{b}. \tag{8}$$

Since the inspection distance is short ($\approx$ 0.08–0.10 m), a worst-case 2° rotational misalignment induces a lateral projection error below 3.5 mm—smaller than the observed seam width and comparable to per-frame detection uncertainty—so the rigid mounting assumption does not dominate the global reconstruction error.

This transformation is applied incrementally to accumulate a globally referenced seam trajectory, governed by three distance thresholds: (i) pose updates are accepted only for displacements exceeding 0.05 m; (ii) seam samples are recorded every $d_{\mathrm{sample}} = 0.05$ m for uniform spatial density; and (iii) displacements exceeding $d_{\mathrm{gap}} = 0.2$ m initialize a new segment to prevent erroneous continuity. The final output is a globally aligned 3D seam curve of discrete, motion-segmented samples; no global structural optimization is applied at this stage.

C. Module 3: Seam-Consistent Synthetic Point Cloud Generation

The globally referenced seam points from Module 2 accumulate incrementally and may lack spatial monotonicity due to motion segmentation; moreover, forward and return passes over the same seam produce two spatially proximate but distinct trajectories that are not automatically merged. Module 3 recovers the geometrically consistent ordering of the seam points—a prerequisite for the sequential marker annotation in Module 4—and generates a synthetic point cloud for operator visualization. The current pipeline does not fuse multi-pass trajectories. Forward and return paths remain separate traces, so the synthetic output reflects single-pass accuracy. Future work will extend the pipeline with multi-pass fusion.

Geometric Ordering. PCA is applied to the accumulated seam centroids $\mathbf{P} \in \mathbb{R}^{N\times 3}$ to extract the dominant eigenvector $\mathbf{v}_1$ of $\mathrm{Cov}(\mathbf{P})$, while points are sorted by projection $\tau_i = (\mathbf{p}_i - \bar{\mathbf{p}})^{\top}\mathbf{v}_1$ to obtain a monotonic sequence along the seam direction.

Trajectory Smoothing. A Savitzky–Golay filter [17] (L=11, order p=3) is applied per axis to suppress per-frame noise while preserving global curve geometry.

Synthetic Cloud Generation. The smoothed trajectory is resampled at arc-length intervals of $\Delta d = 2$ mm, and dense points are generated as:

$$\begin{aligned} \mathbf{p}_{\text{dense}} &= \mathbf{p}_{\text{int}} + \boldsymbol{\epsilon}, \\ \boldsymbol{\epsilon} &\sim \mathcal{U}\big(-\tfrac{\Delta w_{\text{seam}}}{2},\ \tfrac{\Delta w_{\text{seam}}}{2}\big)^3, \end{aligned} \tag{9}$$

where Δw_{seam} denotes the estimated seam width from Module 1. The resulting representation serves as a visual annotation layer over the global map—a synthetic geometric overlay parameterized by the estimated seam width, not a direct surface reconstruction.

D. Module 4: Target-Point Detection and Localization

The final module detects and localizes target points along the reconstructed seam path by combining learning-based visual detection with post-processing-based adaptive clustering and seam-constrained geometric localization. Potential target points are detected from the RGB stream using a YOLO segmentation model [18] fine-tuned on images. For each detection with confidence exceeding a threshold $\tau_{\text{conf}} = 0.60$, the bounding box center (u_c, v_c) is extracted and converted to a metric offset relative to the image center:

$$\Delta x = (u_c - c_x)\rho, \quad \Delta y = (v_c - c_y)\rho, \tag{10}$$

where ρ denotes the metric resolution in meters per pixel, computed from the horizontal field of view θ_{fov} and the working distance d_w as:

$$\rho = \frac{2\, d_w \tan(\theta_{\text{fov}}/2)}{W}, \tag{11}$$

where W is the image width in pixels. The resulting body-frame offset $\mathbf{o}_{\text{target}} = [0,\ -\Delta x,\ -\Delta y]^\top$ is then transformed into the global frame using the robot pose, following the coordinate fusion principle of Module 2:

$$\mathbf{p}^W_{\text{target}} = \mathbf{p}^W_{\text{robot}} + \mathbf{R}^W_b\, \mathbf{o}_{\text{target}}, \tag{12}$$

where $\mathbf{R}^W_b$ is derived from the robot orientation quaternion $\mathbf{q}$. Each detection is projected onto the reconstructed seam path via a KD-tree [19] nearest-neighbor query, yielding an arc-length coordinate s along the seam; detections whose projection distance exceeds a threshold d_{snap} are rejected as off-seam false positives.

Rather than performing online frame-to-frame tracking, all valid detections are stored as raw observations containing 3D position, confidence, timestamp, and robot velocity. These observations are then processed jointly after the run, reducing sensitivity to fixed tracking thresholds and enabling global optimization over the complete observation set.

After collection, the raw detections are consolidated into unique target-point annotations via adaptive clustering on the arc-length domain: detections are sorted by arc-length s, and a sliding-window scan groups consecutive detections into clusters using the adaptive merge threshold

$$d_{\text{merge}} = \min\left(d_{\text{base}} + \lambda \bar{v},\ d_{\max}\right), \tag{13}$$

where $\bar{v} = (v_i + v_{i+1})/2$ is the average robot velocity between adjacent detections, λ is a velocity scaling factor, and $d_{\max}$ is the upper bound on the merge threshold. The parameters are set as $d_{\text{base}} = 0.10\,\text{m}$, $\lambda = 0.5\,\text{s}$, $d_{\max} = 0.30\,\text{m}$, and $n_{\min} = 3$ detections. A new cluster is initiated whenever the arc-length gap exceeds d_{merge}: faster motion produces spatially broader detection distributions for the same physical target point (camera motion blur, reduced temporal overlap), so the threshold widens, while at lower speeds it tightens to prevent merging distinct nearby target points.

After clustering, each group $\mathcal{C}_k = \{(\mathbf{p}_j, s_j, c_j)\}_{j=1}^{N_k}$ is merged into a single target-point annotation using confidence-weighted averaging:

$$\hat{s}_k = \sum_{j=1}^{N_k} w_j\, s_j, \qquad \hat{\mathbf{p}}_k = \sum_{j=1}^{N_k} w_j\, \mathbf{p}_j, \tag{14}$$

where $w_j = c_j / \sum_l c_l$ are the normalized confidence weights. Clusters with fewer than $n_{\min}$ detections are discarded to suppress isolated false positives, and the merged position $\hat{\mathbf{p}}_k$ is snapped to the nearest point on the seam path for geometric consistency with the weld trajectory.

The final output is a set of globally localized target-point annotations (Fig. 2), each defined by its 3D position on the weld seam and arc-length coordinate, published as RViz markers for real-time visualization. Decoupling visual detection from geometric localization allows the detection model to be replaced or extended to new target categories (e.g., cracks) without modifying the clustering or localization pipeline, while velocity-adaptive clustering eliminates manual threshold tuning across inspection speeds.

IV. Experimental Results

Experiments were conducted on a real industrial pressure vessel with both longitudinal and circumferential weld seams under representative industrial operating conditions (Fig. 4). The robot was manually teleoperated along the seam; all perception and mapping modules ran fully autonomously onboard. Each seam geometry (Fig. 4-A) was evaluated with five runs, yielding ten sequences. The evaluation was conducted at two levels: (i) per-frame seam localization accuracy (Modules 1–2), and (ii) global along-seam consistency and target-point localization (Modules 3–4). In all experiments, the pipeline ran at 4.6–4.8 Hz (210–218 ms per frame).

A. Per-Frame Localization Accuracy Evaluation

1) Ground-Truth Annotation: A randomly selected subset of 100 RGB–depth frame pairs (50 per seam geometry) was manually annotated by a trained operator for quantitative evaluation. To quantify intra-operator consistency, the same subset was re-annotated after a two-day interval; the resulting repeatability of 0.422 ± 0.278 mm serves as an empirical lower bound on achievable per-frame accuracy.

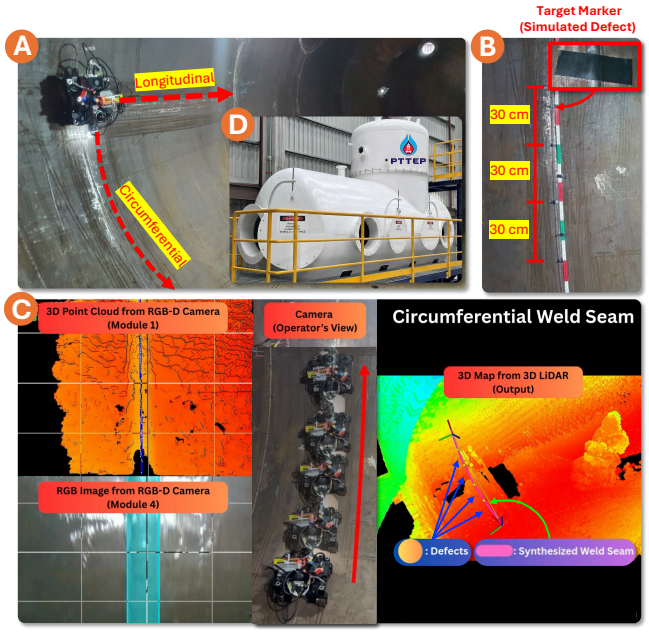


Fig. 4. **Experimental setup and results.** (**A**) Experimental setup inside the pressure vessel showing the two weld seam types: longitudinal and circumferential, indicated by red dashed arrows. (**B**) Target-point markers were placed at 30 cm intervals along the weld seams to simulate defect locations. (**C**) System output during inspection: RGB and depth views from the onboard camera, and the reconstructed 3D global map showing the FAST-LIO2 point cloud with the seam-consistent synthetic point cloud $\mathbf{p}_{\text{dense}}$ (pink colored path) and detected defect markers (yellow sphere) $\hat{\mathbf{p}}_k$ annotated on the weld seam. This result can be seen at https://youtu.be/vpQ86zmROuk. (**D**) Industrial pressure vessel utilized for experimental validation.

2) Evaluation Metrics: Geometric fidelity was quantified by centerline MAE (the primary metric throughout the evaluation), 95th percentile error (P95), and angular error; F1-score and frame-level coverage ratio (Cov.) report detection reliability.

3) Baseline Comparison: A separate held-out set of 300 annotated frames (150 per geometry) was reserved for baseline evaluation, with all parameters fixed prior to testing. Table II compares the proposed depth-based method against minimum-depth (Min-Z) selection and gradient-based edge detectors on depth (Canny-depth) and RGB images (Canny-RGB) [20]—canonical depth-extrema and photometric strategies for curvilinear structure extraction. Note that learning-based baselines [4, 9, 11, 12, 21, 22] were excluded from this comparison, as they lack metric, depth-based centerline output and no public in-vessel weld dataset exists for unbiased retraining.

As shown in Table II, our proposed method achieved lower geometric error and substantially higher success rates. Gradient-based approaches exhibited unstable extraction on curved metallic surfaces, yielding low valid-detection rates; their favorable angular errors, computed over a small surviving subset, did not reflect overall robustness.

TABLE II
BASELINE COMPARISON ON 300 HELD-OUT ANNOTATED FRAMES

Method	MAE (mm)↓	Angular Error (°)↓	Success (%)↑
Min-Z	1.13	1.60	55.7
Canny-depth	0.83	26.9	7.0
Canny-RGB	1.03	2.5	7.5
Ours	**0.87**	**1.4**	**74.6**

TABLE III
ABLATION STUDY

	MAE (mm)↓		P95 (mm)↓		F1↑	
Configuration	C	L	C	L	C	L
Full pipeline	0.87	**0.76**	1.29	**1.16**	**0.650**	**0.676**
− Plane alignment	1.64	1.31	1.82	1.50	0.057	0.149
− RANSAC line fit	1.13	0.90	1.74	1.82	0.574	0.500
− Matched filtering	0.88	0.84	1.23	1.26	0.599	0.475
− Sub-pixel refinement	**0.83**	0.97	**1.10**	1.41	0.634	0.466

C: circumferential, L: longitudinal. −: removes one component from the full pipeline.

4) Per-Geometry Breakdown: Table I reports per-frame results by seam geometry, averaged over five runs each. Longitudinal seams yielded an approximately 30% lower MAE than circumferential ones (0.76 vs. 1.10 mm), which is consistent with their simpler, planar geometry. Conversely, the angular error was lower on circumferential seams (0.90° vs. 1.61°), indicating that lateral localization and direction-estimation errors are not directly coupled across seam geometries. Detection reliability (F1 and Cov.) remained comparable, indicating generalization without geometry-specific tuning.

5) Ablation Study: Table III presents the impact of removing individual components, evaluated on the same 300-frame held-out set as Table II, for circumferential (C) and longitudinal (L) seams. **Plane alignment** was the most critical: without it, the circumferential MAE increased from 0.87 to 1.64 mm and the F1-score dropped below 0.15, indicating that raw depth profiles required surface normalization. Removing **RANSAC line fitting** mainly increased large-error outliers (P95 0.4–0.6 mm), with a smaller effect on MAE. **Matched filtering** provided a modest but consistent improvement across all metrics, while **Sub-pixel refinement** yielded only marginal gains, suggesting that near-pixel accuracy was largely achieved by the preceding stages.

B. Global Seam Trajectory and Target-Point Localization

The seam-constrained reconstruction (Modules 1–2) achieved a global map-level deviation of 13.59 mm MAE (P95 = 28.87 mm) and a Frechet distance of 43.31 mm [23], computed against a reference polyline along the physical seam centerline measured with a flexible tape. This metric reflected accumulated odometry drift and global alignment consistency, rather than per-frame lateral extraction accuracy. With Module 3, the mean deviation decreased from 13.59 to 13.37 mm and the tail error from P95 28.87 to 26.06 mm,

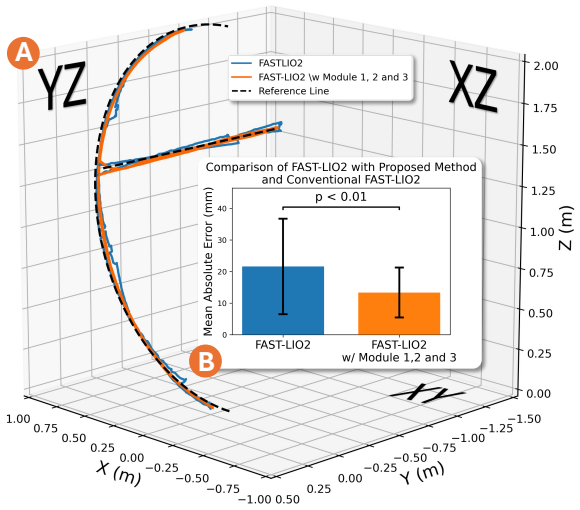


Fig. 5. **MAE comparison of global seam reconstruction.** (**A**) Raw FAST-LIO2 trajectory (blue) compared with the compensated seam path (orange) of the vessel (see Fig. 4), obtained by fusing lateral seam offsets with robot poses and applying ordering and Savitzky–Golay smoothing (Modules 1–3). The resulting trajectory forms a single geometrically consistent seam curve, improving visual coherence and interpretability. (**B**) The progressive reduction in MAE indicates that incorporating seam-based geometric constraints enhances global consistency while improving trajectory continuity and clarity.

indicating improved geometric consistency at the global map scale (Fig. 5).

For target-point localization, visual markers placed at 30 cm intervals along the seam were measured via flexible tape aligned with the weld centerline; each detected marker was projected onto the reconstructed 3D seam curve, and arc-length deviation computed as $e_i = |\hat{s}_i - s_i^{\mathrm{gt}}|$.

Across all runs, the marker localization MAE was 2.60 cm with 96.7% ordering consistency ($\approx$ 8.7% of the 30 cm spacing). This precision is sufficient to re-identify reference points at this interval and support defect re-localization across inspection visits. In the next development phase, we will validate the system performance against task-specific repair tolerances.

V. Discussion

The experimental results validate several key design choices. Per-frame localization accuracy of 0.76 mm (longitudinal) and 1.10 mm (circumferential) is achieved without any learning-based seam detector (Table I). These sub-millimeter errors are within a factor of three relative to human annotation repeatability (0.42 mm, Section IV-A1), suggesting that the deterministic pipeline captures most of the geometrically extractable information in a single RGB-D frame. The global trajectory MAE of 13.59 mm reflects accumulated odometry drift rather than detection inaccuracy, confirming the value of fusing high-rate local measurements with LiDAR–inertial odometry in GPS-denied environments. A first-order error budget supports this: per-frame detection (0.76–1.10 mm) and extrinsic misalignment (<3.5 mm) account for less than a third of the 13.37 mm global MAE, leaving odometry drift as the dominant error source, while formal covariance propagation remains an open issue for further investigation.

TABLE IV
Comparison of Robotic Systems for Weld-Seam Perception and Spatial Representation

Method	Robot	SF	SD	SL	TM	SV
[4]	Mobile	RGB	✓	○	✗	✗
[13]	Mobile	RGB	✓	○	✗	✗
[10]	Mobile	RGB-D	✓	○	✗	✗
[11]	Mobile	RGB, UT	✓	○	✗	✗
[9]	Mobile	RGB-D	✓	○	✗	✗
[12]	Mobile	RGB	✓	○	✗	✗
Ours	**Mobile**	**RGB-D**	✓	●	✓	✓

SF: Sensory Feedback; SD: Seam Detection; SL: Seam Localization; TM: Target Marking; SV: Spatial Visualization; ○: Seam localization performed in image space or robot-centric coordinates without globally consistent spatial registration; ●: Seam localization with globally consistent spatial registration; ✗ indicates the feature is not available or not reported.

Target-point localization achieves 2.60 cm MAE with 96.7% ordering consistency relative to 30 cm intervals, enabled by velocity-adaptive clustering that prevents over- or under-segmentation at varying speeds. The ablation study confirms plane alignment as the single most critical component: its removal nearly doubles the circumferential MAE and reduces F1 below 0.15, validating that geometric normalization must precede feature extraction on arbitrarily oriented surfaces.

Table IV compares our system to related robotic systems, which perform seam tracking in image space or robot-centric coordinates without globally consistent spatial registration [4, 9–13]. The key distinction of the proposed system is the explicit fusion of per-frame measurements with LiDAR–inertial odometry, producing globally referenced seam trajectories that bridge perception and maintenance planning, along with spatial visualization and target marking.

Despite these capabilities, several limitations should be noted. First, the target-point experiments use artificial markers; evaluation on real weld flaws is deferred to future work. Second, the Intel RealSense D405 is a general-purpose sensor, not compliant with API 510 [1] inspection-grade requirements. Third, all experiments were conducted on one industrial vessel under controlled conditions—generalization to corroded, coated, or non-standard weld profiles has not been evaluated. Finally, the 4.8 Hz frame rate is adequate for teleoperation but insufficient for closed-loop operation at higher speeds.

VI. Conclusion and Future Work

This paper presents a framework for magnetic-wheel inspection robots to perform weld-seam detection and spatial localization, integrating deterministic depth-based extraction, odometry fusion, and velocity-adaptive target-point clustering to produce centimeter-level seam maps. Future work will focus on deploying inspection-grade sensing compliant with API 510, performing multi-vessel validation across 0.5–3 m

diameters with diverse surface conditions and weld geometries (including non-cylindrical heads), benchmarking against learning-based seam detectors trained on our expanded dataset, and evaluating performance on real weld defects such as porosity, undercut, and cracks. In addition, closed-loop autonomous seam tracking with wheel-slip compensation, sustained 15–20 Hz operation on embedded platforms, and seam-constrained pose graph optimization will be pursued to enhance robustness and deployment readiness.

ACKNOWLEDGMENTS

This research received funding support from the National Science, Research and Innovation Fund (NSRF) via the Program Management Unit for Human Resources & Institutional Development, Research and Innovation [grant number B13F680099]. It was also supported by PTT Exploration and Production Public Company Limited through the HERO2.1 project.

REFERENCES

[1] American Petroleum Institute, *API 510: Pressure Vessel Inspection Code*, 11th ed. API, 2022.

[2] X. Li, Y. Wei, Z. Lv *et al.*, "High resolution weld semantic defect detection algorithm based on integrated double u structure," *Scientific Reports*, vol. 15, p. 17849, 2025.

[3] D. Schmidt and K. Berns, "Climbing robots for maintenance and inspections," *Robotics and Autonomous Systems*, 2013.

[4] J. Wang, L. Huang, J. Yao, M. Liu, Y. Du, M. Zhao, Y. Su, and D. Lu, "Weld seam tracking and detection robot based on artificial intelligence technology," *Sensors*, vol. 23, no. 15, p. 6725, 2023.

[5] W. Xu, Y. Cai, D. He, J. Lin, and F. Zhang, "FAST-LIO2: Fast direct LiDAR-Inertial odometry," *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2053–2073, 2022.

[6] T. Shan, B. Englot, D. Meyers, W. Wang, C. Ratti, and D. Rus, "LIO-SAM: Tightly-coupled lidar inertial odometry via smoothing and mapping," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 5135–5142.

[7] I. Khaled, D. Vasiukov, M. Shakoor, M. Bennebach, and S. Chaki, "Digital twin for predicting progressive damage in operating pressure vessels," *Procedia Structural Integrity*, vol. 57, pp. 280–289, 2024.

[8] *NIIMax: 3D Workflow Software for Non-Intrusive Inspection*, Quasset and ESR Technology, 2022.

[9] J. Li, C. Tu, F. Xu, and X. Wang, "Weld seam recognition and tracking with omnidirectional wall-climbing robot for spherical tank inspection," *ASME Journal of Autonomous Vehicles and Systems*, vol. 5, no. 1, p. 011002, 2025.

[10] M. Zhong, M. Pan, Z. Mao, R. Lyu, and Y. Liu, "A wall-climbing robot with a mechanical arm for weld inspection of large pressure vessels," *Actuators*, vol. 14, no. 12, p. 607, 2025.

[11] J. Wang, J. Yao, X. Liu, Y. Du, M. Liu, Y. Su, and D. Lu, "Weld detection and tracking algorithm for inspection robot based on deep learning," 01 2024, pp. 85–92.

[12] M. Zhao, X. Liu, K. Wang, Z. Liu, Q. Dong, P. Wang, and Y. Su, "Welding seam tracking and inspection robot based on improved yolov8s-seg model," *Sensors*, vol. 24, no. 14, p. 4690, 2024.

[13] M. Zhang, X. Zhang, M. Li, J. Cao, and Z. Huang, "Optimization design and flexible detection method of a surface adaptation wall-climbing robot with multisensor integration for petrochemical tanks," *Sensors*, vol. 20, no. 22, p. 6651, 2020.

[14] M. A. Fischler and R. C. Bolles, "Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography," *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, 1981.

[15] O. Rodrigues, "Des lois géométriques qui régissent les dź placements d'un système solide dans l'espace, et de la détermination des variations élémentaires des orbites," *Journal de Mathématiques Pures et Appliquées*, vol. 5, pp. 380–440, 1840.

[16] C. Steger, "An unbiased detector of curvilinear structures," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 20, no. 2, pp. 109–125, 1998.

[17] A. Savitzky and M. J. E. Golay, "Smoothing and differentiation of data by simplified least squares procedures." *Analytical Chemistry*, vol. 36, no. 8, pp. 1627–1639, 1964.

[18] R. Sapkota and M. Karkee, "YOLOE-26: Integrating YOLO26 with YOLOE for real-time open-vocabulary instance segmentation," 01 2026.

[19] J. Friedman, J. Bentley, and R. Finkel, "An algorithm for finding best matches in logarithmic expected time," *ACM Trans. Math. Softw.*, vol. 3, pp. 209–226, 09 1977.

[20] J. Canny, "A computational approach to edge detection," *Pattern Analysis and Machine Intelligence, IEEE Transactions on*, vol. PAMI-8, pp. 679 – 698, 12 1986.

[21] J. Li, C. Tu, F. Xu, and X. Wang, "Weld seam recognition and tracking with omnidirectional wall-climbing robot for spherical tank inspection," *Journal of Autonomous Vehicles and Systems*, vol. 5, pp. 1–31, 10 2024.

[22] J. Wang, L. Huang, J. Yao, M. Liu, Y. Du, M. Zhao, Y. Su, and D. Lu, "Weld seam tracking and detection robot based on artificial intelligence technology," *Sensors*, vol. 23, no. 15, p. 6725, 2023.

[23] M. M. Fréchet, "Sur quelques points du calcul fonctionnel," *Rendiconti del Circolo Matematico di Palermo (1884-1940)*, vol. 22, no. 1, pp. 1–72, 1906.